

# A Practical Guide to Mixture model

TOPIC -- MIXTURE MODEL

-- 01 --

**Extensions** In a Bayesian setting, additional levels can be added to the graphical model defining the mixture model. For example, in the common latent Dirichlet allocation topic model, the observations are sets of words drawn from  $D$  different documents and the  $K$  mixture components represent topics that are shared across documents. Each document has a different set of mixture weights, which specify the topics prevalent in that document. All sets of mixture weights share common hyperparameters. A very common extension is to connect the latent variables defining the mixture component identities into a Markov chain, instead of assuming that they are independent identically distributed random variables. The resulting model is termed a hidden Markov model and is one of the most common sequential hierarchical models. Numerous extensions of hidden Markov models have been developed; see the resulting article for more information.

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Generate  $N$  random numbers from a categorical distribution of size  $n$  and probabilities  $p_i$  for  $i = 1$  to  $n$ . These tell you which of the  $F_i$  each of the  $N$  values will come from. Denote by  $m_i$  the quantity of random numbers assigned to the  $i$ th category. For each  $i$ , generate  $m_i$  random numbers from the  $F_i$  distribution.

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**Point set registration** Probabilistic mixture models such as Gaussian mixture models (GMM) are used to resolve point set registration problems in image processing and computer vision fields. For pair-wise point set registration, one point set is regarded as the centroids of mixture models, and the other point set is regarded as data points (observations). State-of-the-art methods are e.g. coherent point drift (CPD) and Student's  $t$ -distribution mixture models (TMM). The result of recent research demonstrate the superiority of hybrid mixture models (e.g. combining Student's  $t$ -distribution and Watson distribution/Bingham distribution to model spatial positions and axes orientations separately) compare to CPD and TMM, in terms of inherent robustness, accuracy and discriminative capacity.

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$$K = \{ p(\cdot) : p(\cdot) = \sum_{i=1}^n a_i f_i(\cdot; \theta_i), a_i > 0, \sum_{i=1}^n a_i = 1, f_i(\cdot; \theta_i) \in \mathcal{J} \forall i, n \}$$

$;\theta_{i}), a_i > 0, \sum_{i=1}^n a_i = 1, f_i(\cdot; \theta_i) \in \mathcal{J} \setminus \forall \text{for all } i, n \right\}$

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$$\mu_{s^{(j+1)}} = \frac{\sum_{t=1}^N h_{s^{(j)}}(t) x(t)}{\sum_{t=1}^N h_{s^{(j)}}(t)} \quad \{\displaystyle \mu_{s^{(j+1)}} = \frac{\sum_{t=1}^N h_{s^{(j)}}(t) x(t)}{\sum_{t=1}^N h_{s^{(j)}}(t)}\}$$

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Other methods Some of them can even probably learn mixtures of heavy-tailed distributions including those with infinite variance (see links to papers below). In this setting, EM based methods would not work, since the Expectation step would diverge due to presence of outliers.

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$N$  random variables that are observed, each distributed according to a mixture of  $K$  components, with the components belonging to the same parametric family of distributions (e.g., all normal, all Zipfian, etc.) but with different parameters. However, it is also possible to have a finite mixture model where each component belongs to a different parametric family of distributions, for example, a mixture of a multivariate normal distribution and a generalized hyperbolic distribution.  $N$  random latent variables specifying the identity of the mixture component of each observation, each distributed according to a  $K$ -dimensional categorical distribution A set of  $K$  mixture weights, which are probabilities that sum to 1. A set of  $K$  parameters, each specifying the parameter of the corresponding mixture component. In many cases, each "parameter" is actually a set of parameters. For example, if the mixture components are Gaussian distributions, there will be a mean and variance for each component. If the mixture components are categorical distributions (e.g., when each observation is a token from a finite alphabet of size  $V$ ), there will be a vector of  $V$  probabilities summing to 1. In addition, in a Bayesian setting, the mixture weights and parameters will themselves be random variables, and prior distributions will be placed over the variables. In such a case, the weights are typically viewed as a  $K$ -dimensional random vector drawn from a Dirichlet distribution (the conjugate prior of the categorical distribution), and the parameters will be distributed according to their respective conjugate priors. Mathematically, a basic parametric mixture model can be described as follows:

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Parameter estimation and system identification Parametric mixture models are often used when we know the distribution  $Y$  and we can sample from  $X$ , but we would like to determine the  $\mu_i$  and  $\sigma_i$  values. Such situations can arise in studies in which we sample from a population that is composed of several distinct subpopulations. It is common to think of probability mixture modeling as a missing data problem. One way to understand this is to assume that the data points under consideration have "membership" in one of the distributions we are using to model the data. When we start, this membership is unknown, or missing. The job of estimation is to devise appropriate parameters for the model functions we choose, with the connection to the data points being represented as their membership in the individual model distributions. A variety of approaches to the problem of mixture decomposition have been proposed, many of which focus on maximum likelihood methods such as expectation maximization (EM) or maximum a posteriori estimation (MAP). Generally these methods consider separately the questions of system identification and parameter estimation; methods to determine the number and functional form of components within a mixture are distinguished from methods to estimate the corresponding parameter values. Some notable departures are the graphical methods as outlined in Tarter and Lock and more recently minimum message length (MML) techniques such as Figueiredo and Jain and to some extent the moment matching pattern analysis routines suggested by McWilliam and Loh (2009).

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House prices Assume that we observe the prices of  $N$  different houses. Different types of houses in different neighborhoods will have vastly different prices, but the price of a particular type of house in a particular neighborhood (e.g., three-bedroom house in moderately upscale neighborhood) will tend to cluster fairly closely around the mean. One possible model of such prices would be to assume that the prices are accurately described by a mixture model with  $K$  different components, each distributed as a normal distribution with unknown mean and variance, with each component specifying a particular combination of house type/neighborhood. Fitting this model to observed prices, e.g., using the expectation-maximization algorithm, would tend to cluster the prices according to house type/neighborhood and reveal the spread of prices in each type/neighborhood. (Note that for values such as prices or incomes that are guaranteed to be positive and which tend to grow exponentially, a log-normal distribution might actually be a better model than a normal distribution.)